Power Budget Example

Team Number:	209	
Project Name:	**Auto-Lock Door Sensor**	
Team Member Names:	**Matthew Sanderson, Andrew Imoukhuede, Dylan Vierra-Guillermo, Bryce Weber**	
Version:	**1**	

All Major Components	**Component Name**	**Part Number**	**Supply Voltage Range**	**#**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	74HC165 Shift Register	74HC165	+2 - 6V	1	60	60	mA
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.1V	1	500	500	mA
	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA
+5V Power Rail	**Component Name**	**Part Number**	**Supply Voltage Range**	**#**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	74HC165 Shift Register	74HC165	+2 - 6V	1	60	60	mA
	Curiosity Nano	PIC18F57Q43	+1.8 - 5.1V	1	500	500	mA
					Subtotal	560	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	700	mA
Regulator	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA
					Total Remaining Current Available on +5V Rail	700	mA
External Power Source 1	**Component Name**	**Part Number**	**Supply Voltage Range**	**Output Voltage**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
Power Source 1 Selection	Plug-in Wall Supply	(full part number)	110VAC	+9V	3000	5000	mA
Power Rails Connected to External Power Source 1	AOT2618L MOSFET	AOT2618L TO-220	-20V - +20V	1	1400	1400	mA
					Total Remaining Current Available on External Power Source 1	3600	mA